BARINDER SINGH BANWAIT

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ABOUT

Research mathematician and software engineer working on Lean 4 formalisation and machine learning for number theory. My Lean work includes a complete proof of the Ramanujan–Nagell theorem, and statements of open conjectures, including the Birch–Swinnerton-Dyer conjecture, for Google DeepMind's *Formal Conjectures* benchmark for automated theorem provers. My machine-learning papers train neural networks and decision trees on large datasets of elliptic curves; in the most recent, the models led to new formulae, which we then proved.

BA and MMath (Cambridge), PhD in mathematics (Warwick), and academic positions including INRIA Bordeaux, MIT, and Boston University. Industry experience in surgical robotics and quantitative finance, with production code in PyTorch, Python, and safety-critical C. Long-standing contributor to open-source mathematical software; led a research group of five from proposal to publication.

SKILLS

Machine Learning & AI

PyTorch | TensorFlow | scikit-learn | pandas | NumPy | Multi-agent AI workflows for mathematical research

Programming Languages & Tooling

Python | Lean 4 / Mathlib | C/C++ | Sage | Magma | PARI/GP | Git | AWS | Postgres

PAPERS AND PREPRINTS

(authors are listed alphabetically, following the convention in mathematics)

Machine learning for mathematics

1. **Decision trees, Frobenius traces, and Weierstrass coefficients of elliptic curves**, with X. Huang, K.-H. Lee, S. Lee, T. Oliver and A. Pozdnyakov. Submitted, *arXiv:2607.24251* (2026).
2. **Machine Learning Approaches to the Shafarevich-Tate Group of Elliptic Curves**, with A. Babei, A. Fong, X. Huang and D. Singh. *Advances in Theoretical and Mathematical Physics*. 29(8) 2353-2379 (2025).

Formal mathematics

3. **Second derivatives of p-adic L-functions and the Shafarevich–Tate group of rank-two CM elliptic curves**. Submitted, *arXiv:2609.08431* (2026). (Includes a Lean 4 formalisation of the main criterion, modulo cited results.)
4. **A formal proof of the Ramanujan–Nagell theorem in Lean 4**. Submitted, *arXiv:2604.09808* (2026).

Number theory and arithmetic geometry

5. **On the Visibility category of the Shafarevich–Tate group**, with J. Caro and S. Chidambaram. Submitted, *arXiv:2601.21519* (2026).
6. **On the Identification of Elliptic Curves That Admit Infinitely Many Twists Satisfying the Birch–Swinnerton-Dyer Conjecture**, with X. Huang. To appear, *Algorithmic Number Theory Symposium XVII* (2026).
7. **Torsion subgroups of elliptic curves over quadratic fields and a conjecture of Granville**, with M. Derickx. *Algorithmic Number Theory Symposium XVI* (2024).
8. **Towards strong uniformity for isogenies of prime degree**, with M. Derickx. Submitted, *arXiv:2302.08350* (2024).
9. **Computing nonsurjective primes associated to Galois representations of genus 2 curves**, with A. Brumer, H. J. Kim, Z. Klagsbrun, J. Mayle, P. Srinivasan and I. Vogt. *Contemporary Mathematics 796* (2023).
10. **Modularity over $\mathbb{C}$ implies modularity over $\mathbb{Q}$**. To appear, *Contemporary Mathematics* (2023). (Proceedings of *Modularity and the generalized Fermat equation*.)
11. **Explicit isogenies of prime degree over number fields**, with M. Derickx. *Algebra and Number Theory*, 19(6), 1147–1197 (2025).
12. **Cyclic isogenies of elliptic curves over fixed quadratic fields**, with F. Najman and O. Padurariu. *Mathematics of Computation*. Volume 93, 841–862 (2023).
13. **Explicit isogenies of prime degree over quadratic fields**. *International Mathematics Research Notices*. 2023(14):11829–11876 (2023).
14. **Examples of abelian surfaces failing the local-global principle for isogenies**. *Research in Number Theory*. 7(55) (2021)
15. **Del Pezzo surfaces over finite fields and their Frobenius traces**, with F. Fité and D. Loughran. *Mathematical Proceedings of the Cambridge Philosophical Society*. 167(1) (2019) 35-60.

16. **Tetrahedral Elliptic Curves and the local-global principle for isogenies**, with J. Cremona. *Algebra and Number Theory.* 8:5 (2014) 1201-1229.
17. **On some local to global phenomena for abelian varieties**. PhD Thesis, University of Warwick (2013).

OPEN SOURCE SOFTWARE CONTRIBUTIONS



Google DeepMind *Formal Conjectures* - Lean 4, Mathlib

2026–present

Open benchmark of research-level problems stated in Lean 4 (arXiv:2605.13171), for evaluating automated theorem provers and autoformalisation systems. Merged contributions:

- The weak Birch–Swinnerton-Dyer conjecture over number fields, with definitions of the L-function and minimal discriminant of an elliptic curve (#5322, #5371, #5548); the strong form is in progress.
- Goldfeld's conjecture on ranks in families of quadratic twists, in three variants (#5173).
- The elliptic curve over $\mathbb{Q}$ of rank at least 31 found in August 2026 (#5159).

Formalizing the Ramanujan–Nagell Theorem - Lean 4, Mathlib4

2026

Formally verifying that the equation $x^2 + 7 = 2^n$ has only 10 integer solutions (x, n).

L-Functions and Modular Forms Database - Python, Postgres, MongoDB

2020–2025

Data management and UI across Elliptic Curves, Classical and Bianchi Modular Forms, Testing utilities, Dirichlet Characters, and Modular Curves.

Absolutely simple endomorphism rings - SageMath, Python

2021

First functionality to check for geometric simplicity of Jacobians of genus 2 curves over $\mathbb{Q}$. Released in sage-9.5.

RESEARCH LEADERSHIP AND MENTORING

Raising a Mathematician Workshop

Feb 2026 — Mumbai, India

Led a group of students through the open congruent number problem in lectures and problem sessions. Working with Mumbai-based *Raising a Mathematician* foundation to promote passion for mathematical education in India.

Effective Algebra and the LMFDB

Jan 2025 — Kampala, Uganda

Led the research project *L-functions and Galois representations* at this school of the *Centre International de Mathématiques Pures et Appliquées* (CIMPA), funded by the French government to promote mathematical research in developing countries.

Rethinking Number Theory 5

June 2024 — Online

Invited to lead the project *Machine-Learning the rank and Shafarevich-Tate group of a rational elliptic curve* at this American Institute of Mathematics research community. Proposed the project and led a group of five to publication (paper 2 above).

SELECTED TALKS

(the link goes to my slides or the video of me giving the talk)

- **Realizing cubic isogeny primes: AI-assisted experiments**. *Using AI to find examples*, CRM Montréal (2026).
- **On the Visibility category of the Shafarevich–Tate group**. *Algebraic Points on Curves*, ICERM, Brown University (2025).
- **Cyclic isogenies of elliptic curves over fixed quadratic fields**. *Modern Breakthroughs in Diophantine Problems*, BIRS (2022).
- **Modularity over $\mathbb{C}$ implies modularity over $\mathbb{Q}$**. *The Generalized Fermat Equation*, Bhaskaracharya Pratishthana (2022).
- **Explicit isogenies of prime degree over quadratic fields**. *VaNTAGe Seminar* (2021).

INDUSTRY EXPERIENCE

Research Engineer

CMR Surgical

Jan 2018 – Sep 2019 | Cambridge, UK

- Implementing machine learning algorithms for robot arm condition monitoring, using PyTorch and TensorFlow.
- Time-series telemetry processing of sensor data in Python, using pandas, numpy, and matplotlib.
- Research and optimisation of robotic control algorithms, including inverse kinematics and mass-spring-damper models.
- Writing production-level, safety-critical embedded C code, compliant with MISRA C and International Standard IEC 62304.
- Unit and regression tests (Google Test) under continuous integration; data pipelines on AWS (Lambda, S3, Athena).

Quantitative Analyst

Quantile

Sep 2019 – Mar 2020 | London, UK

Linear, mixed-integer, and multi-objective optimisation (Gurobi) for compression of interest-rate derivative portfolios.

ACADEMIC POSITIONS

Lodha Mathematical Sciences Institute

Nov 2025 – Mar 2026 | Mumbai, India

Massachusetts Institute of Technology

Sep 2024 – Aug 2025 | Cambridge, MA, USA

Boston University

Sep 2022 – Aug 2025 | Boston, MA, USA

Ruprecht-Karls-Universität Heidelberg

Oct 2021 – Apr 2022 | Heidelberg, Germany

Harish-Chandra Research Institute

Feb – Sep 2021 | Prayagraj, India

Max-Planck-Institut für Mathematik

Oct 2016 – Nov 2016 | Bonn, Germany

Universität Duisburg-Essen

Jan 2015 – Jan 2017 | Essen, Germany

Institut national de recherche en informatique et en automatique (INRIA)

Jan – Dec 2014 | Bordeaux, France

EDUCATION

PhD Mathematics

University of Warwick

Jan 2010 – Sep 2013 | Coventry, UK

Supervisor: Prof. John Cremona

BA and MMath Mathematics

University of Cambridge - Christ's College

Oct 2005 – June 2009 | Cambridge, UK

MMath (*Part III of the Mathematical Tripos*) - Distinction

LANGUAGES

English | ਪੰਜਾਬੀ (Punjabi) | Deutsch | हिंदी (Hindi) | Français